zika — Extracted Transmission Models

Overview

A total of 169 transmission models were extracted from 93 articles that were flagged for containing extractable transmission models. The predominant model architecture was Compartmental (79.3%), with Vector/Animal to human (66.9%) representing the most commonly modelled transmission route.

Summary

Metric	Value
Models extracted	169
Articles considered	93
Deterministic models	94 (55.6%)
Stochastic models	55 (32.5%)
Models with available code	35 (20.7%)

Figure 1: Model Overview

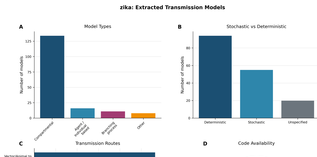

Figure 1. Overview of 169 transmission models extracted for zika. (A) Distribution of model architectures. (B) Classification by stochastic or deterministic formulation. (C) Primary transmission routes incorporated. (D) Proportion of models with publicly available source code.

Model Formulation

Deterministic formulations were employed in 55.6% of models, while stochastic approaches accounted for 32.5%. Empirical data were incorporated in 36.1% of the extracted models, with 55.6% classified as purely theoretical.

Figure 2: Interventions and Assumptions

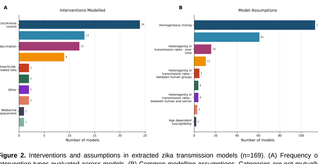

Figure 2. Interventions and assumptions in extracted zika transmission models (n=169). (A) Frequency of intervention types evaluated across models. (B) Common modelling assumptions. Categories are not mutually exclusive; individual models may incorporate multiple interventions or assumptions.

Interventions and Assumptions

The most frequently modelled interventions were Vector/Animal control (14.2%), Treatment (7.7%), Vaccination (7.1%). Common modelling assumptions included Homogeneous mixing (66.9%), Heterogeneity in transmission rates - over time (9.5%), Heterogeneity in transmission rates - between groups (6.5%).

Figure 3: Programming Languages

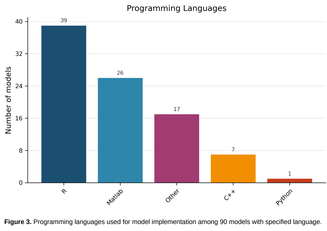

Figure 3. Programming languages used for model implementation among 90 models with specified language.

Figure 4: Model Type by Formulation

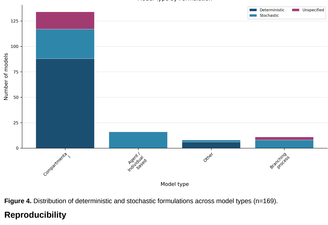

Figure 4. Distribution of deterministic and stochastic formulations across model types (n=169).

Reproducibility

Source code was publicly available for 20.7% of the extracted models.

Tables

Table 1: Model Types

Model Type	Count	Proportion
Compartmental	134	79.3%
Agent / Individual based	16	9.5%
Branching process	11	6.5%
Other	8	4.7%

Distribution of model architectures across 169 extracted transmission models.

Table 2: Model Formulation

Formulation	Count	Proportion
Deterministic	94	55.6%
Stochastic	55	32.5%
Unspecified	20	11.8%

Classification of extracted models by stochastic versus deterministic formulation (n=169).

Table 3: Transmission Routes

Transmission Route	Count	Proportion
Vector/Animal to human	113	66.9%
Vector/Animal to human;Sexual	21	12.4%
Unspecified	14	8.3%
Sexual	6	3.6%
Vector/Animal to human;Sexual;Human to human (direct contact)	3	1.8%
Vector/Animal to human;Human to human (direct contact)	3	1.8%
Human to human (direct contact);Vector/Animal to human;Sexual	2	1.2%
Human to human (direct contact)	2	1.2%
Vector/Animal to human;Sexual;Human to human (direct non-sexual contact)	2	1.2%
Vector/Animal to human;Human to human (direct non-sexual contact)	1	0.6%
Vector/Animal to human;Human to human (direct non-sexual contact);Sexual	1	0.6%
Sexual;Vector/Animal to human	1	0.6%

Primary transmission routes incorporated in extracted models (n=169).

Table 4: Interventions Modelled

Intervention Type	Count	Proportion
Vector/Animal control	24	14.2%
Treatment	13	7.7%
Vaccination	12	7.1%
Behaviour changes	9	5.3%
Insecticide-treated nets	2	1.2%
Pesticides/larvicides	2	1.2%
Other	2	1.2%
Hospitals	2	1.2%
Wolbachia replacement	1	0.6%
Genetically modified mosquitoes	1	0.6%

Types of interventions evaluated in extracted models. Categories are not mutually exclusive (n=169).

Table 5: Model Assumptions

Assumption	Count	Proportion
Homogeneous mixing	113	66.9%
Other	61	36.1%
Heterogeneity in transmission rates - over time	16	9.5%
Heterogeneity in transmission rates - between groups	11	6.5%
Heterogeneity in transmission rates - between human groups	5	3.0%
Latent period is same as incubation period	4	2.4%
Heterogeneity in transmission rates - between human and vector	4	2.4%
Cross-immunity between Zika and dengue	3	1.8%
Age dependent susceptibility	2	1.2%

Common modelling assumptions. Categories are not mutually exclusive (n=169).

Table 6: Code Availability

Code Available	Count	Proportion
Yes	35	20.7%
No	132	78.1%

Availability of publicly accessible source code for extracted models (n=169).

Table 7: Programming Languages

Programming Language	Count	Proportion
Unspecified	79	46.7%
R	39	23.1%
Matlab	26	15.4%
Other	17	10.1%
C++	7	4.1%
Python	1	0.6%

Programming languages used for model implementation (n=169).

Table 8: Empirical Data Usage

Empirical Data Used	Count	Proportion
Yes	61	36.1%
No/Unspecified	108	63.9%

Model Inventory (Sample)

Table 9 presents a sample of 10 extracted model records. The complete dataset of 169 extracted transmission models for zika is publicly available.

Table 9: Sample of Extracted Models

Article ID	Model Type	Compartmental Structure	Formulation	Transmission Route	Spatial Scale	Co
PMID_28095405	Compartmental	SEIR-SEI	Unspecified	Vector/Animal to human	False	Fa
PMID_28933344	Agent / Individual based	Not compartmental	Stochastic	Vector/Animal to human	Unspecified	Fa
PMID_28933344	Agent / Individual based	Not compartmental	Stochastic	Vector/Animal to human	Unspecified	Fa
PMID_28933344	Agent / Individual based	Not compartmental	Stochastic	Vector/Animal to human	Unspecified	Fa
PMID_28933344	Agent / Individual based	Not compartmental	Stochastic	Vector/Animal to human	Unspecified	Fa
PMID_28933344	Agent / Individual based	Not compartmental	Stochastic	Vector/Animal to human	Unspecified	Fa
PMID_27774987	Compartmental	Other compartmental	Deterministic	Vector/Animal to human	Unspecified	Fa
PMID_30133450	Compartmental	SEIR-SEI	Deterministic	Vector/Animal to human	Unspecified	Fa
PMID_30133450	Other	Not compartmental	Deterministic	Vector/Animal to human	Unspecified	Fa
PMID_30133450	Other	Not compartmental	Deterministic	Vector/Animal to human	Unspecified	Fa

Sample of 10 records from 169 extracted zika transmission models.

Data Availability

The complete dataset of extracted transmission models for zika (n=169) has been made publicly available to support future modelling efforts and outbreak preparedness.